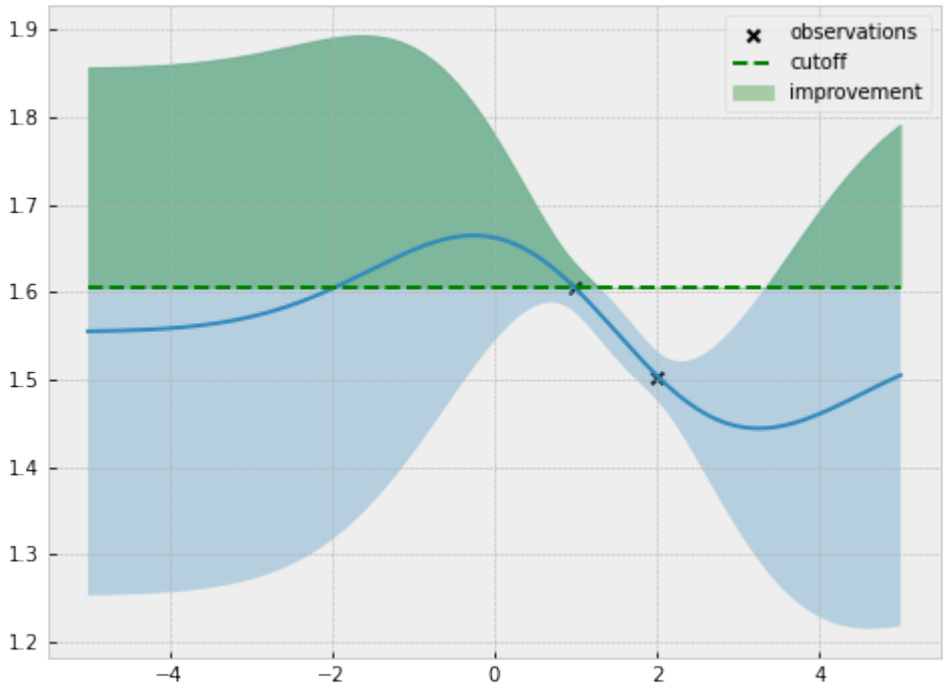


Figure 4.8 Improvement from the incumbent from the perspective of a Gaussian process. The portion of the Gaussian process' predictions that corresponds to improvement from the incumbent is highlighted darker.

Reasoning about the probability that $x = 0$ will yield a higher function value is easy to do, as by querying point $x = 0$, the improvement from the incumbent that we may observe exactly corresponds to what's called a normal distribution that is partially truncated, as shown in Figure 4.9.

The left panel of Figure 4.9 contains the same GP in Figure 4.8 that is cut off at the incumbent, with an addition showing the CI of the normal distribution prediction at $x = 0$. Slicing the GP vertically at this point 0, we obtain the right panel of Figure 4.9, where the CIs in the two panels are the same. We see that only the highlighted portion of the normal distribution in the right panel represents the improvement we may observe from the incumbent, which is what we care about. This highlighted portion is part of a normal distribution, which as we will see in the upcoming subsections, leads to many mathematical conveniences.

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